



WGNI FARM

STANDARD OPERATING PROCEDURE

GREENHOUSE OPERATIONS

FIELD	DETAIL
Document Title	Greenhouse Operations Standard Operating Procedure
Covers	Bell Pepper Production - All Active Greenhouse Kits including Nursery Kit
Location	WGNI Farm, Abeokuta, Nigeria
Document Authority	CEO, WGNI Farm
Applies To	Greenhouse Worker, Agronomist, Farm Manager
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DEFINITIONS

Fertigation: The process of delivering dissolved nutrients to plants through the drip irrigation system simultaneously with water. Plants receive water and nutrition in a single, precisely controlled delivery through each drip emitter.

pH: A measure of the acidity or alkalinity of the nutrient solution on a scale of 0 to 14. Values below 7 are acidic and above 7 are alkaline. Bell pepper requires a pH of 5.8 to 6.5 for optimal nutrient uptake. Outside this range, nutrients become unavailable to the plant even if they are present in the solution.

EC (Electrical Conductivity): A measure of the concentration of dissolved nutrients in the irrigation solution, expressed in millisiemens per centimetre (mS/cm). EC below the target range means plants are underfed. EC above the target range causes root damage and wilting.

Mulch Film: Black and white polythene sheeting laid over raised planting beds to suppress weeds, retain soil moisture, regulate soil temperature, and reduce soil-borne disease contact with the plant. Plants are established through holes cut directly into the mulch film.

Drip Emitter: The individual outlet point on a drip irrigation line through which water and nutrient solution is delivered directly to each plant at the base of the stem.

Foliar Feeding: The application of dissolved nutrients directly onto the leaf surface as a spray. The plant absorbs the nutrients through the leaf tissue. Foliar feeding is used to supplement root-applied nutrition, correct deficiencies rapidly, or support the crop during periods of high demand.

Blossom Drop: The premature falling of flowers before they develop into fruit, typically caused by heat stress, humidity imbalance, or poor pollination conditions.

Damping Off: A fungal disease condition that causes seedlings to collapse at soil level, characterised by a soft rotted stem at the base. It is most common in conditions of excessive moisture and poor air circulation.

Interveinal Chlorosis: Yellowing of leaf tissue between the veins while the veins themselves remain green. It indicates a specific nutrient deficiency, and the affected nutrient is identified by which leaves show symptoms first.

Fruit Set: The stage at which a successfully pollinated flower begins to develop into a fruit. Poor fruit set means flowers are dropping without forming fruit.

Re-entry Interval: The minimum time that must pass after a chemical application before any person may safely re-enter the greenhouse, as specified on the product label. This interval must be strictly observed without exception.

Anthrachnose: A fungal disease that causes sunken dark circular spots on fruits. It is one of the most common post-flowering diseases in bell pepper cultivation and spreads rapidly in wet or humid conditions.

SECTION 1 - ACCESS AND SECURITY

1.1 Authorised Personnel

Only the following personnel are permitted to enter any active greenhouse kit:

- The assigned Greenhouse Worker
- The Agronomist
- The Farm Manager

No other person may enter under any circumstance. This restriction applies without exception to farm workers from other units, visitors, buyers, contractors, and any other individual regardless of seniority or relationship to the farm.

1.2 Entry Protocol

Every authorised person entering a greenhouse kit must observe the following without exception:

- Remove or cover external footwear at the entrance. Dedicated greenhouse footwear assigned to each kit must be worn inside at all times.
- Wash hands thoroughly with soap and water before touching any plant, planting bed, irrigation line, or tool.
- Close and secure the door immediately upon entering. The door must remain closed and secured at all times while anyone is inside. An open door allows uncontrolled pest entry and disrupts the internal growing environment.
- When moving from one kit to another, wash hands and clean or change footwear before entering the next kit. Secure the door of the vacated kit before proceeding to the next.

1.3 Kit Separation

Each greenhouse kit is an independent growing unit. Tools, footwear, and equipment assigned to one kit must not be transferred to another without cleaning and disinfection using Hypo solution. If one kit is under active pest or disease management, movement of personnel and equipment between kits is suspended until the Agronomist confirms it is safe to resume.

1.4 Visitor Protocol

Visitors and buyers observe from outside the greenhouse entrance only. If the Farm Manager grants permission for a visitor to enter, that visitor must comply fully with Section 1.2 including the immediate closure of the door on entry. No visitor may touch any plant, planting bed, irrigation line, or equipment.

1.5 Security

- All greenhouse kits must be secured when no authorised personnel are present.
- No plant material, produce, equipment, or chemical may leave the greenhouse without authorisation from the Agronomist or Farm Manager.
- Any evidence of unauthorised entry, tampering, or theft must be reported to the Farm Manager immediately.

SECTION 2 - ENVIRONMENTAL AND PERIMETER CONDITIONS

2.1 Perimeter Clearance Standard

A minimum clearance zone of 2 metres must be maintained around the exterior of every active greenhouse kit at all times. Within this zone:

- Grass must not exceed 5 centimetres in height at any time.
- All weeds must be removed at the root.
- No organic debris including cut grass, fallen leaves, or dead vegetation may accumulate.
- No standing water may be present at any time.
- No materials including equipment, bags, or chemical containers may be stored within the zone.

The gravelled drainage channel running along the base of the greenhouse exterior wall must be kept clear of debris, silt, and vegetation at all times to ensure effective drainage.

The Farm Manager inspects the perimeter during the weekly Saturday inspection. Any perimeter not meeting this standard must be corrected before the following working day.

2.2 Internal Floor and Walkway Conditions

The following standards apply inside every greenhouse kit at all times:

- Walkways must be clear, firm, and free of loose soil, plant debris, fallen fruits, chemical spills, and standing water.
- No grass or weed growth is permitted inside any kit. Any growth observed must be removed by hand immediately. Herbicides must never be used inside an active greenhouse.
- The base of the greenhouse structure where the polythene side wall meets the ground must be checked weekly for gaps. Any gap must be sealed or reported to the Farm Manager immediately.
- Soil or growing medium displaced from the planting beds onto walkways must be collected and returned or removed immediately.

2.3 Mulch Film Condition

The mulch film covering all planting beds must be inspected during every observation walkthrough. Report to the Agronomist:

- Any tear or perforation in the mulch film beyond the designated planting holes.
- Mulch film that has lifted, shifted, or detached from the bed edge.
- Planting holes that have widened excessively, exposing bare soil around the plant stem.
- Any section of mulch film that has become waterlogged, sagging, or pooling water on its surface.

Damaged mulch film compromises weed suppression, moisture retention, and disease control. All damage is reported on the day it is observed and repaired or replaced as the Agronomist directs.

2.4 Ventilation

The greenhouse kits at WGNI Farm operate without mechanical vents. Ventilation is passive through the shade netting on the roof and the open ends of the greenhouse structure. The Agronomist manages ventilation by controlling the open end panels and shade netting position as required. The Greenhouse Worker does not adjust or close the open ends of the greenhouse without Agronomist instruction.

2.5 Algae

Green algae growth on any surface inside the greenhouse including walkways, the greenhouse structure, or the mulch film must be reported to the Agronomist on the day it is observed. Affected surfaces are cleaned with dilute Hypo solution (50ml to 75ml Hypo in 10 to 15 litres of clean water) or as directed by the Agronomist. The underlying moisture or air circulation cause must be identified and corrected.

SECTION 3 - WATER SUPPLY AND PUMPING

3.1 System

Water is transferred from the main storage tank to the greenhouse header tank using a 2HP mud pump located at the main tank. From the header tank, water flows through the drip irrigation lines to the plants during each fertigation run.

3.2 Who Operates the Pump

The mud pump may be operated by:

- The Greenhouse Worker
- The Agronomist
- The Farm Manager

3.3 When to Pump

- Check the header tank level every morning before irrigation and every evening before the evening fertigation run.
- Pump from the main tank whenever the header tank drops below half capacity.
- Never allow the header tank to run completely empty. An empty header tank introduces air into the drip lines causing uneven water and nutrient delivery across the entire kit.

3.4 Pumping Procedure

1. Confirm the main storage tank has sufficient water before starting the pump.
2. Open the valve connecting the main tank to the pump delivery line.
3. Start the pump.
4. Remain present while the pump is running. A running pump must never be left unattended.
5. Monitor the header tank level as it rises.
6. Switch the pump off when the header tank reaches full capacity. Do not overfill.
7. Close the valve.
8. Confirm the pump is fully off before leaving the area.
9. Record the pump run on Form A: time started and time stopped.

3.5 Fault Identification and Response

Table 3.1 - Pump Fault Response Guide

Fault Observed	Immediate Action
Abnormal pump sounds: rattling, grinding, or high-pitched noise	Switch off immediately. Report to Farm Manager.
Water leaking from delivery pipe during pumping	Switch off. Report to Farm Manager.
Pump running but header tank level not rising	Switch off immediately. Report to Farm Manager.
Main storage tank empty	Do not start the pump. Report to Farm Manager immediately.

3.6 Pump Rules

- Never run the pump with the main tank empty.
- No repair or adjustment to the pump is to be carried out without Farm Manager authorisation.
- The pump must be switched off and the valve closed after every session without exception.

SECTION 4 - FERTIGATION AND IRRIGATION

4.1 Schedule

Fertigation runs twice daily: morning and evening. The morning run begins after the Agronomist has confirmed pH and EC and after the worker has completed the morning observation walkthrough and confirmed drip line integrity. The evening run begins after the evening observation walkthrough is completed. The duration of each run is determined by the Agronomist. The Greenhouse Worker does not alter timing or duration without Agronomist instruction.

4.2 Pre-Run Checks by the Agronomist

Before every fertigation run the Agronomist must:

- Measure and record the pH of the nutrient solution. Target: 5.8 to 6.5. Adjust before the run if outside this range.
- Measure and record the EC of the nutrient solution. Target: 2.0 to 3.5 mS/cm. Adjust before the run if outside this range.
- Confirm the nutrient solution is correctly prepared for the current growth stage.
- Confirm the fertigation injector is correctly connected and set to the prescribed injection rate.
- Record all pre-run parameters on the Notion app before the run begins.

4.3 Nutrient Preparation

- All nutrient solutions are prepared by the Agronomist or under direct Agronomist supervision.
- The Greenhouse Worker does not mix, dilute, or modify any nutrient solution without the Agronomist present.
- Nutrient concentrates are stored in the designated locked chemical store and issued by the Agronomist per mixing session.

4.4 During the Irrigation Run

- Walk every row during every run. Do not monitor from a fixed position.
- Confirm every drip emitter is delivering water. A non-dripping emitter indicates blockage or disconnection and must be reported to the Agronomist immediately.
- Record the location of any pooling on the mulch film surface and report to the Agronomist.
- Record the location of any dry bed section adjacent to wet sections and address the emitter immediately.
- If the fertigation injector is not drawing solution at the correct rate, stop the run and report to the Agronomist before proceeding.

4.5 After Each Run

- Close all system valves properly.
- At the end of the evening run only, flush drip lines with clean water for a minimum of 2 to 3 minutes to clear residual nutrient solution and prevent salt accumulation and emitter blockage.
- Record on Form A: kit number, start time, end time, nutrient solution details, and any observations.
- The Agronomist records post-run observations on the Notion app.

4.6 Drip Line and Emitter Maintenance

- Inspect all drip lines and emitters before every irrigation session.
- Clear a blocked emitter by removing it, rinsing it in clean water, and reinstalling. If the blockage does not clear, replace the emitter and record the fault on Form A.
- Report cracked, split, or disconnected lines to the Agronomist. No improvised repair is permitted without Agronomist instruction.
- Never displace or pull plants to access drip lines.

4.7 pH Adjustment Protocol

- pH above 6.5: lower using dilute phosphoric acid or nitric acid solution. Add in small measured increments and re-measure after each addition.

- pH below 5.8: raise using potassium hydroxide or calcium hydroxide solution. Add in small measured increments and re-measure after each addition.
- Re-check pH 10 minutes after any adjustment before initiating the fertigation run.
- Record every pH adjustment on Form B: additive used, volume added, starting pH, and final pH.

4.8 EC Adjustment Protocol

- EC below target: increase nutrient concentrate in small measured increments. Re-measure before running fertigation.
- EC above target: dilute with clean water in measured increments. Re-measure before running fertigation.
- A sudden EC increase without any change in the nutrient mix may indicate salt accumulation in the planting beds. Flush affected bed sections with clean water and monitor closely. Report to the Farm Manager.

SECTION 5 - FOLIAR NUTRITION

5.1 What Foliar Feeding Is

Foliar feeding is the application of dissolved nutrients directly onto the leaf surface as a fine spray. The plant absorbs the nutrients through the leaf tissue rapidly, making foliar application faster in effect than soil or irrigation-applied nutrition. It is used to supplement root-applied fertigation, correct visible deficiencies quickly, or support the crop during stages of high nutritional demand such as flowering and fruit development.

Foliar feeding supplements fertigation. It does not replace it. Both must continue as scheduled.

5.2 When to Apply Foliar Feed

- During the vegetative stage to support rapid leaf and stem growth when the root system is still developing.
- At the onset of flowering to boost potassium and calcium availability at the point of flower formation.
- When visible nutrient deficiency symptoms are observed and rapid correction is required. See Section 14.3.
- During periods of heat stress when root uptake is reduced and foliar delivery provides a direct and immediate supply.
- As directed by the Agronomist based on crop observation and the nutrient programme schedule.

5.3 Common Foliar Formulations

The Agronomist selects the appropriate formulation based on the growth stage and crop condition. Common formulations used in greenhouse bell pepper production include:

Table 5.1 - Common Foliar Nutrient Formulations

Formulation Type	Nutrient Profile	Primary Use Stage	Notes
Balanced NPK (e.g. 20:20:20 or similar)	Equal Nitrogen, Phosphorus, Potassium	Vegetative and establishment	General growth support. Use when no specific deficiency is targeted.
High-Potassium soluble (e.g. 9:18:36 or similar)	Low N, low P, high K	Flowering and fruit development	Supports fruit set, colour development, and shelf life.
Calcium-Magnesium supplement	Calcium and Magnesium	Flowering and fruiting	Prevents blossom end rot and interveinal chlorosis. Critical during fruit swelling.
Iron chelate solution	Chelated Iron (Fe-EDTA or Fe-DTPA)	Any stage when iron deficiency observed	Applied to young leaves showing interveinal chlorosis. Do not mix with phosphate solutions.
Boron supplement	Boron (B)	Pre-flowering and flowering	Supports pollen viability and fruit set. Applied at low concentration - excess causes toxicity.

5.4 Foliar Application Procedure

- Foliar feeds are prepared and supervised by the Agronomist. The worker does not prepare foliar solutions without the Agronomist present.
- Dissolve the foliar fertiliser in clean water at the rate specified by the Agronomist or the product label.
- Apply using a clean knapsack or pressure sprayer. Spray equipment must be thoroughly rinsed before use to ensure no chemical residue from a previous application contaminates the foliar feed solution.
- Spray both the upper and lower surfaces of the leaves until the point of run-off.

- Apply in the early morning before 9:00 AM or in the late afternoon after 4:00 PM. Never apply during the midday period when temperatures are at peak. Heat causes rapid evaporation of the solution from the leaf surface before absorption can occur, and can cause leaf scorch.
- Do not apply foliar feed immediately before or during an irrigation run.
- Do not apply foliar feed on the same day as a pesticide or fungicide spray unless the Agronomist has specifically confirmed compatibility. Incompatible tank mixes can damage the crop and reduce efficacy.

5.5 Foliar Application Frequency

- Foliar feeding is not a daily activity. Frequency is determined by the Agronomist based on the crop's growth stage and observed condition.
- As a general guideline, a balanced foliar feed may be applied once every 7 to 14 days during the vegetative and flowering stages.
- Correction sprays for specific deficiencies are applied as directed by the Agronomist until the deficiency is resolved.
- Every foliar application must be recorded on Form D immediately after completion.

5.6 Safety Rules for Foliar Application

- Wear gloves, nose mask, and long-sleeved clothing during all foliar spray applications.
- Do not eat, drink, or smoke during or immediately after application.
- Wash all exposed skin and hands thoroughly with soap and water after application.
- Return all foliar fertiliser products to the locked chemical store after use.

Rule: No foliar application is made without the Agronomist's direct instruction. Every application is recorded on Form D.

SECTION 6 - DAILY ROUTINES

6.1 Morning Routine (7:00 AM)

- 6.1.1 Arrive at the greenhouse. Put on dedicated footwear. Wash hands.
- 6.1.2 Enter the first kit. Close and secure the door immediately.
- 6.1.3 Conduct the Morning Observation Walkthrough. See Section 7.
- 6.1.4 Check all drip lines and emitters along every row. Record and report all faults to the Agronomist before irrigation begins.
- 6.1.5 Check the header tank water level. Arrange pumping if below half capacity. See Section 3.
- 6.1.6 Wait for Agronomist confirmation that pH, EC, and nutrient solution are ready before starting the irrigation run.
- 6.1.7 Carry out the morning fertigation run. Walk every row during the run. See Section 4.
- 6.1.8 Confirm even water distribution across all plants. Report any plant not receiving water to the Agronomist immediately.
- 6.1.9 Secure the door. Move to the next kit. Repeat 6.1.3 to 6.1.8 for each active kit.
- 6.1.10 Complete morning entries on Form A for each kit.
- 6.1.11 Secure all greenhouse entrances.

6.2 Evening Routine (5:00 PM)

- 6.2.1 Wash hands. Put on greenhouse footwear. Enter the first kit. Close and secure the door immediately.
- 6.2.2 Conduct the Evening Observation Walkthrough. See Section 7.
- 6.2.3 Check all drip lines and emitters. Report new faults to the Agronomist.
- 6.2.4 Check the header tank level. Arrange pumping if insufficient for the following morning.
- 6.2.5 Wait for Agronomist confirmation that pH and EC have been checked before starting the run.
- 6.2.6 Carry out the evening fertigation run. Walk every row during the run.
- 6.2.7 After the run, flush drip lines with clean water for 2 to 3 minutes.
- 6.2.8 Confirm even water distribution across all plants.
- 6.2.9 Secure the door. Move to the next kit. Repeat 6.2.2 to 6.2.8 for each active kit.
- 6.2.10 Complete evening entries on Form A for each kit.
- 6.2.11 Secure all greenhouse entrances. Submit Form A to the Agronomist.

SECTION 7 - OBSERVATION WALKTHROUGH

The observation walkthrough is conducted every morning and every evening in every active kit. Walk every row. Examine every plant. Do not skip rows or check from a fixed position.

7.1 Canopy

Before entering, observe the overall canopy from the entrance. Note any areas where plants appear shorter, paler, or visibly stressed compared to surrounding plants.

7.2 Plant Body - Top to Bottom

Walk each row and examine:

- Leaves for yellowing, browning, spots, curling, or distortion
- Stems for discolouration, softening, or unusual narrowing at any point
- Fruits for spots, cracks, sunken areas, or abnormal colour
- Flowers for premature drop

7.3 Leaf Undersides

Turn leaves over and examine the underside at every session. Check for:

- White powdery residue
- Tiny insects, eggs, or crawling organisms
- Sticky residue indicating insect feeding activity
- Fine webbing between leaves or along stems

7.4 Stem Base

Examine the base of each plant stem at mulch film level. Check for:

- Dark soft rot or discolouration at the base of the stem
- White, grey, or brown fuzzy fungal growth
- Stem collapse or narrowing at soil level
- Widened planting holes in the mulch film exposing bare soil around the stem

7.5 Planting Beds and Mulch Film

Check along every bed for:

- Pooling water on the mulch film surface
- Sections of mulch film that are lifted, torn, or displaced
- Dry areas of the bed adjacent to wet areas
- Green algae growth on the mulch film or bed surface

7.6 Drip Lines and Emitters

As you walk, check for:

- Disconnected or leaking lines
- Dry emitters during or after an irrigation run
- Lines displaced from their correct position at the plant base

7.7 Greenhouse Structure

Check for:

- Tears or openings in the shade netting or side polythene
- Damage to the roof covering
- Gaps at the base of the structure
- Grass or weed growth inside the kit - remove by hand immediately

7.8 Record and Report

Record all observations on Form A. Use Section 8 to determine the appropriate response and timing.

SECTION 8 - OBSERVATION STANDARDS AND RESPONSE GUIDE

Table 8.1 - Plant and System Observation Response Guide

What You See	Likely Cause	Action Required	When
Lower leaves yellowing across multiple plants	Nitrogen deficiency or overwatering	Report to Agronomist. Record row numbers.	Same day
Interveinal chlorosis on older leaves	Magnesium or potassium deficiency	Report to Agronomist	Same day
Interveinal chlorosis on young leaves only	Iron deficiency	Report to Agronomist	Same day
Brown or black spots on leaves with yellow border	Fungal or bacterial infection	Do not remove leaves. Report to Agronomist immediately.	Immediately
White powder on upper leaf surface	Powdery mildew	Report to Agronomist immediately.	Immediately
Wilting despite recent irrigation	Root rot, stem rot, or severe pest damage	Check bed moisture. Report to Agronomist immediately.	Immediately
Distorted or curling new leaves	Aphid, mite, or virus	Check leaf undersides. Report to Agronomist.	Immediately
Sticky residue on leaves with visible insects	Aphid or whitefly infestation	Report to Agronomist. Record rows affected.	Immediately
Fine webbing between leaves or stems	Spider mite infestation	Report to Agronomist immediately.	Immediately
Tiny white insects flying up when plant is disturbed	Whitefly	Report to Agronomist. Record rows affected.	Immediately
Dark soft rot at stem base at mulch film level	Phytophthora or stem rot	Do not irrigate that plant. Report to Agronomist immediately.	Immediately
Sunken dark spots on fruits	Anthrachnose	Report to Agronomist. Do not harvest affected fruits.	Immediately
Blossom drop across multiple plants	Heat stress, humidity imbalance, or nutrient problem	Report to Agronomist. Check structure ends for air movement.	Same day
More than 30 percent of flowers failing to set fruit	Emergency fruit set failure	Report to Agronomist immediately. Farm Manager to be informed same day.	Immediately
Fruits cracking or splitting	Irregular irrigation or calcium deficiency	Report to Agronomist.	Same day
Dry bed section adjacent to wet sections	Blocked emitter or disconnected line	Clear or replace emitter. Report if unresolved.	Immediately

What You See	Likely Cause	Action Required	When
Pooling water on mulch film surface	Overwatering or drainage failure	Report to Agronomist. Withhold next irrigation from that section without instruction.	Same day
Torn, lifted, or displaced mulch film	Physical damage or pest activity	Report to Agronomist. Record location.	Same day
Green algae on mulch film, floor, or structure	Excess moisture or poor air movement	Report to Agronomist.	Same day
Torn netting or structural gaps	Pest entry risk	Report to Farm Manager immediately.	Immediately
Grass or weed growth inside the kit	Environmental control failure	Remove by hand immediately. Report to Farm Manager.	Immediately

Rule: The Greenhouse Worker observes and reports. No chemical application, plant removal, or system adjustment is made without the Agronomist's direct instruction.

SECTION 9 - NURSERY KIT

9.1 Current Arrangement and Standard

Seedling trays are currently placed on the ground within an active growing kit. This is an interim arrangement only. The standard for WGNI Farm is that all nursery trays are elevated on benches or tables at a minimum height of 80 centimetres from the ground. Trays placed directly on bare ground are exposed to soil-borne pathogens, waterlogging, and uncontrolled pest access, all of which compromise seedling health and transplant success rate.

When the dedicated nursery kit is commissioned, it will operate under the full access, cleaning, environmental, and record-keeping standards of this SOP. Until then, the nursery area within the current kit must observe the same standards as the growing area.

9.2 Tray Preparation

- Wash and disinfect all trays with Hypo solution (50ml to 75ml Hypo in 10 to 15 litres of clean water) before every use.
- Fill trays with the growing medium as directed by the Agronomist. No external or unsterilised soil is used in nursery trays.
- Sow seeds at the depth and spacing directed by the Agronomist, typically 1 to 1.5 cm deep, one seed per cell.

9.3 Daily Nursery Care

- Check all seedling trays every morning and evening.
- Keep the growing medium consistently moist. Water gently using a fine rose watering head. Do not flood cells.
- Report to the Agronomist any seedling that has not germinated 10 to 14 days after sowing.
- Remove any seedling showing signs of collapse at the base immediately and report to the Agronomist.

9.4 Nursery Observation Guide

Table 9.1 - Nursery Observation Guide

What You See	What It Indicates	Action Required
Seedling collapsed at soil level	Damping off disease	Remove immediately. Report to Agronomist.
Seedling pale or uniformly yellow	Nutrient deficiency or inadequate light	Report to Agronomist.
Seedling leaning strongly to one side	Uneven light distribution	Report to Agronomist. Rotate tray if instructed.
Spots or mould on seedling leaves	Fungal infection	Report to Agronomist immediately.
No germination after 14 days	Poor seed viability or adverse conditions	Report to Agronomist immediately.
Tray waterlogged or sitting in pooled water	Ground placement drainage failure	Elevate tray immediately. Report to Agronomist.

9.5 Transplant Readiness Criteria

The Agronomist determines when seedlings are ready for transplanting. Standard indicators:

- 4 to 6 true leaves fully developed (not counting the seed leaves)
- Plant height of 10 to 15 cm
- Stem is firm and upright
- No visible pest or disease activity

- Root system visible at the base of the cell but not severely pot-bound
- Approximately 4 to 6 weeks from sowing date

9.6 Nursery Cleanliness

- Remove all dead or failed seedlings from trays daily.
- Keep the nursery floor or bench surface clean and dry at all times.
- Disinfect all trays and tools between production batches using Hypo solution.
- No old trays, empty containers, or debris may accumulate in the nursery area.

SECTION 10 - GROWTH STAGE SCHEDULES

The bell pepper crop passes through six defined growth stages from transplanting to end of cycle. The tasks listed under each stage represent the minimum daily and weekly operational requirements for that period. The Agronomist may add to or adjust these tasks based on crop condition at any time.

10.1 TRANSPLANTING STAGE - Week 1

Daily Tasks

- Water each transplanted seedling gently after planting and at every subsequent morning and evening session.
- Walk every row and inspect every plant. Report any wilting, fallen, or collapsed plant to the Agronomist immediately.
- Confirm the drip emitter is correctly positioned at the base of each plant stem through the mulch film hole.
- Check that mulch film holes have not widened or lifted around new transplants. Report any exposed soil around the stem base.
- Remove any failed transplant and report to the Agronomist.
- Remove all displaced soil and debris from walkways and the mulch film surface.

Weekly Tasks

- Record the plant count for each kit on Form A.
- Confirm all drip lines are correctly routed along each bed.
- Confirm all mulch film edges are secured along the bed length.
- Full walkway and bed surface clean at the end of the week.

10.2 ESTABLISHMENT STAGE - Week 2 to Week 3

During this stage the plant is developing its root system into the bed. Visible above-ground growth is slow. This is normal and expected.

Daily Tasks

- Conduct the full observation walkthrough every morning and evening.
- Confirm every emitter is dripping and every plant is receiving water.
- Inspect the base of every stem at mulch film level for dark or soft rot. Report immediately to the Agronomist if observed.
- Remove all plant debris from the walkways and mulch film surface.
- Report any plant still showing wilting 3 days after transplanting.

Weekly Tasks

- Check bed moisture in multiple locations. The bed should feel consistently moist, not wet and not dry.
- Clean drip line filters.
- Wipe down the greenhouse structure to remove dust and debris.
- Inspect all mulch film edges for lifting or displacement.
- Record all plant losses for the week on Form A.

10.3 VEGETATIVE STAGE - Week 4 to Week 7

The plant is growing rapidly. Staking, binding wire installation, and the first pruning session begin during this stage.

Daily Tasks

- Conduct the full observation walkthrough morning and evening.
- Check drip lines and emitters at every session.
- Inspect every plant for signs of pest activity, particularly on leaf undersides. See Section 11.
- Remove all fallen leaves and plant debris from walkways and the mulch film surface after every session.
- Report any plant growing significantly slower or faster than surrounding plants to the Agronomist.
- Check that all stakes are firmly in position and no plant has fallen.
- Check mulch film condition along all beds.

Weekly Tasks

- Agronomist conducts pruning inspection and carries out pruning as required. See Section 10 (Pruning).
- Check and adjust binding wire on all plants as growth advances.
- Record plant population count and any losses on Form A.
- Flush drip lines at the end of the week.
- Full Saturday clean.
- Agronomist reviews pH and EC records and adjusts nutrient programme as required.

10.4 FLOWERING STAGE - Week 7 to Week 10

Flowers are developing. The quality of management at this stage directly determines the fruit yield of the entire cycle.

Daily Tasks

- Conduct the full observation walkthrough morning and evening.
- Observe and record the number of flowers visible per row. A sudden reduction must be reported to the Agronomist same day.
- Monitor for blossom drop across multiple plants. Report to Agronomist same day if observed.
- Check that the open ends of the greenhouse are managed for adequate air movement. Stagnant humid air during flowering encourages Botrytis.
- Check drip lines and emitters at every session.
- Remove all debris from walkways and mulch film surface.
- Do not touch or disturb flowers unless specifically instructed by the Agronomist.

Weekly Tasks

- Agronomist assesses pollination quality and fruit set rate.
- Agronomist adjusts nutrient programme to increase potassium and calcium at this stage.
- Check and adjust binding wire. Plant mass is increasing significantly.
- Inspect closely for thrips and aphids. Report any sign to the Agronomist immediately.
- Full Saturday clean.
- Record all blossom drop observations on Form A with row number and date.

Flowering Stage Standards

- Daytime temperature must remain between 21 and 27 degrees Celsius. Temperatures above 30 degrees Celsius cause rapid flower abortion.
- Relative humidity must remain between 60 and 70 percent during flowering. Humidity above 75 percent encourages Botrytis grey mould which attacks and destroys flowers directly.

- Chemical applications must not be made during peak flower opening. Schedule sprays for early morning before 9:00 AM or late afternoon after 4:00 PM only.
- If more than 30 percent of flowers are failing to set fruit, this is an emergency condition. The Agronomist must be notified immediately and the Farm Manager informed the same day.

10.5 FRUIT DEVELOPMENT STAGE - Week 10 to Week 14

Fruit is actively developing and increasing in size. Consistent water and nutrient delivery is critical throughout this stage.

Daily Tasks

- Conduct the full observation walkthrough morning and evening.
- Inspect every fruit. Report spots, cracks, sunken areas, or abnormal colour to the Agronomist immediately.
- Monitor for blossom end rot, a dark sunken patch at the base tip of the fruit indicating calcium deficiency. Report immediately.
- Check drip lines and emitters at every session.
- Remove all debris from walkways and mulch film surface after every session.

Weekly Tasks

- Agronomist assesses fruit development against expected size benchmarks.
- Check and reinforce binding wire across all plants. Developing fruits significantly increase plant load.
- Record estimated fruit count per kit on Form A.
- Full Saturday clean.
- Agronomist reviews and adjusts pH and EC as required.

10.6 HARVEST STAGE - Week 14 Onwards

Fruits have reached maturity. Harvesting occurs every 2 to 3 days and continues until the Agronomist determines the cycle is ending.

Daily Tasks

- Conduct the full observation walkthrough morning and evening.
- Assess fruits for harvest readiness. See Section 12.
- After each harvest, inspect for damaged or diseased fruits remaining on the plant. Remove and present to the Agronomist before disposal.
- Check drip lines and emitters at every session.
- Remove all fallen or discarded fruits from walkways and the mulch film surface immediately after harvesting.

Weekly Tasks

- Record total harvest by weight and grade on Form E at every harvest session.
- Assess overall plant condition and vigour. Report any significant decline to the Agronomist.
- Report to the Agronomist if the number of new fruits forming is declining significantly.
- Full Saturday clean.
- Agronomist reviews crop performance and advises on continuation or cycle termination.

SECTION 11 - PRUNING AND PLANT TRAINING

11.1 What Pruning Is

Pruning is the selective removal of specific plant parts including unwanted stems, lateral shoots (suckers), leaves, and excess fruits to direct the plant's energy toward productive fruit development, improve air circulation within the canopy, and reduce disease pressure. In bell pepper cultivation, pruning is an ongoing management activity throughout the vegetative and fruiting stages of the crop cycle.

11.2 Who Carries Out Pruning

Pruning is carried out by the Agronomist or by a Greenhouse Worker under direct Agronomist supervision. No worker prunes independently.

11.3 When Pruning Begins and Continues

- First pruning session: approximately Week 4 to Week 5 of the vegetative stage.
- Subsequent pruning: the Agronomist assesses plants weekly and prunes as required throughout the remainder of the crop cycle.

11.4 What Is Removed During Pruning

Table 11.1 - Pruning Reference Guide

Plant Part Removed	Reason for Removal
Suckers below the first fork	Divert energy from the productive main stem structure
Leaves in contact with the mulch film surface	Direct contact with the mulch introduces disease risk at the stem base
Dead or yellowing leaves	Source of fungal sporulation and disease spread within the canopy
Weak thin non-productive stems	Do not support quality fruit and reduce air circulation
Excess fruits beyond 2 at a single node	Reduces competition allowing remaining fruits to develop to full size

11.5 Pruning Procedure

- Use clean sharp scissors or a pruning knife only.
- Disinfect the cutting tool before moving to each new plant using Hypo solution (1 part Hypo to 9 parts water). Dip the tool and allow 30 seconds contact before use on the next plant.
- Make a clean cut close to the stem. Do not leave extended stubs.
- Never tear or pull plant material off by hand.

11.6 Post-Pruning

- Collect all removed plant material from the walkways and mulch film surface immediately after pruning.
- Place in a sealed bag and remove from the greenhouse completely.
- Pruned material must not remain inside the greenhouse overnight.
- Record all pruning events on Form A: date, kit number, and description of work carried out.

11.7 Binding Wire - Installation and Management

Binding wire or soft twine supports the bell pepper plant vertically as it grows, preventing stem collapse under the increasing weight of developing foliage and fruits.

When First Installed: Binding wire is first installed when plants reach 25 to 30 cm in height, typically around Week 3 to Week 4 of the vegetative stage. Overhead support wires or horizontal frames are installed by the Agronomist before the crop requires them.

How to Attach Binding Wire

1. Cut a length of soft twine as directed by the Agronomist.
2. Tie one end to the overhead support wire or horizontal frame above the plant row.
3. Loop the twine loosely around the main stem just below a leaf junction. The loop must accommodate two fingers between the twine and the stem. It must never be tight against the stem.
4. Twist the twine once between each loop so the plant is held in position and does not slide downward.
5. Continue adding loops upward as the plant grows, adding a new loop every 15 to 20 cm of new growth.

Binding Rules

- Twine must never be tied tightly against the stem. A tight loop restricts vascular flow and causes stem damage as the plant grows.
- Only soft twine is used. Wire, rope, or any material that can cut into the stem is not permitted.
- New loops are added weekly during the vegetative and fruiting stages.
- A plant that has fallen from its support is not forced back into position. Report to the Agronomist immediately.
- All binding is checked weekly. Tight or broken loops are replaced immediately.

SECTION 12 - PEST AND DISEASE CONTROL

12.1 Operational Rules

- No chemical may be applied in any greenhouse kit without the Agronomist's direct authorisation and the Farm Manager's awareness.
- Every chemical application must be recorded on Form D immediately after completion.
- Full protective equipment must be worn during all chemical applications: gloves, nose mask, protective eyewear, and long-sleeved clothing.
- Chemical sprays must not be applied during an irrigation run or immediately before one begins.
- The re-entry interval stated on the product label must be observed after every spray application. The Agronomist confirms the re-entry time before any worker re-enters the kit.

12.2 Pest Identification and Chemical Control

Table 12.1 - Common Pests of Bell Pepper and Recommended Treatments

Pest	Symptoms	Recommended Chemical	Application	Frequency
Whitefly	White insects fly up when plant is disturbed. Sticky film on leaves. Leaf yellowing.	Cypermethrin, Imidacloprid, Thiamethoxam (Actara)	Foliar - undersides of leaves	Every 7 days until controlled. Rotate chemicals.
Aphids	Clusters of soft insects on new growth. Sticky film on leaves. New leaves curling.	Imidacloprid, Dimethoate, Cypermethrin	Foliar - new growth and leaf undersides	Every 7 days until controlled
Spider Mites	Fine webbing between leaves. Bronze or silver leaf discolouration. Tiny organisms on leaf undersides.	Abamectin (Dyneamec), Dicofol, Kelthane	Foliar - both leaf surfaces	Every 5 to 7 days. Rotate at every application.
Thrips	Silver streaks or scarring on leaves and fruits. Tiny slender insects visible in flowers.	Spinosad (Tracer), Imidacloprid, Abamectin	Foliar - leaves and flower area	Every 7 days. Rotate chemicals.
Leaf Miners	Winding pale tunnels visible inside leaf tissue when held to light.	Abamectin (Dyneamec), Cyromazine (Trigard)	Foliar spray	Every 7 to 10 days
Caterpillars and Armyworm	Holes in leaves and fruits. Droppings visible on leaf surfaces.	Cypermethrin, Lambda-cyhalothrin (Karate), Chlorpyrifos	Foliar spray	When observed
Root Knot Nematode	Stunted wilting plant unresponsive to irrigation. Root galls visible when roots examined.	Carbofuran (Furadan) applied to growing medium	Soil drench into planting bed	At planting or as directed by Agronomist

12.3 Disease Identification and Chemical Control

Table 12.2 - Common Diseases of Bell Pepper and Recommended Treatments

Disease	Symptoms	Recommended Chemical	Application	Frequency
Powdery Mildew	White powdery coating on upper leaf surface. Spreads progressively across the canopy.	Sulfur fungicide (Thiovit), Triadimefon (Bayleton), Myclobutanil	Foliar spray	Every 7 to 10 days from first sign
Downy Mildew	Yellow patches on upper leaf. Grey or purple fuzzy growth on underside.	Mancozeb (Dithane M-45), Metalaxyl (Ridomil), Cymoxanil	Foliar spray	Every 7 days especially in humid conditions
Anthrachnose	Sunken dark circular spots on fruits, sometimes with a pink spore mass at the centre.	Mancozeb, Chlorothalonil (Bravo), Carbendazim	Foliar and fruit spray	Every 7 to 10 days
Bacterial Wilt	Sudden total plant wilting. Brown vascular discolouration visible when stem is cut.	No curative chemical. Remove and destroy plant immediately. Drench surrounding soil with Copper Oxychloride.	Soil drench around affected plant	Immediately on detection
Phytophthora Root and Stem Rot	Dark brown or black rot at stem base at mulch film level. Plant collapses.	Metalaxyl (Ridomil Gold), Fosetyl-aluminium (Aliette)	Soil drench at plant base	At first sign. Withhold irrigation from affected plant.
Botrytis Grey Mould	Grey fuzzy mould on stems, leaves, and fruits. Prevalent in cool humid conditions.	Iprodione (Rovral), Carbendazim, Thiram	Foliar spray	Every 7 days during cool or humid periods
Cercospora Leaf Spot	Small circular brown spots with pale centres on leaves. Severe infection causes leaf drop.	Mancozeb, Carbendazim, Copper Oxychloride	Foliar spray	Every 7 to 10 days
Fusarium Wilt	One-sided yellowing and wilting. Brown internal stem discolouration when cut.	No curative chemical. Remove plant. Drench growing medium with Carbendazim.	Soil drench around affected area	Immediately on detection
Damping Off	Seedlings or young transplants collapse at soil level. Stem pinched or rotted at base.	Metalaxyl, Thiram, Captan	Growing medium drench	Preventive application at transplanting stage

12.4 Virus Diseases

Virus diseases including Pepper Mild Mottle Virus, Tobacco Mosaic Virus, and Cucumber Mosaic Virus have no chemical cure. They are transmitted primarily by aphids and thrips.

Symptoms: mosaic yellow and green patterns on leaves, distorted growth, stunted plants, and abnormal colour patterns on fruits.

Protocol:

- Report to the Agronomist immediately on observation.
- Do not handle a suspect plant and then handle healthy plants without washing hands thoroughly between plants.
- The Agronomist determines whether the plant is to be removed and destroyed.
- Prevention is the primary control: maintain aggressive aphid and thrips management at all stages.

12.5 Chemical Rotation

No chemical active ingredient may be applied more than twice consecutively for the same pest or disease. On the third application, a different active ingredient must be used. The Agronomist manages the rotation schedule and records every application on Form D.

12.6 Chemical Safety Rules

- Wear full protective gear during every application: gloves, nose mask, protective eyewear, long-sleeved clothing.
- Never spray against the wind direction.
- Do not eat, drink, or smoke during or immediately after application.
- After application, wash all exposed skin and hands thoroughly with soap and water.
- All chemicals must be returned to the designated locked chemical store immediately after use. Chemicals are never stored inside the greenhouse.
- Empty chemical containers are disposed of as directed by the Agronomist. Containers must never be burned or reused for any other purpose.

SECTION 13 - HARVESTING

13.1 Expected Harvest Timeline

- Green (unripe) harvest: approximately 60 to 70 days from transplanting.
- Coloured (ripe) harvest: approximately 80 to 90 days from transplanting.

The Agronomist confirms the exact harvest commencement date based on observed crop development.

13.2 Harvest Readiness Standards

Table 13.1 - Harvest Readiness Guide

Indicator	Green Harvest	Coloured Harvest
Size	Full size reached. Firm to touch.	Full size. Firm.
Colour	Uniform deep green. No pale or white patches.	Full colour change - red, yellow, or orange depending on variety.
Skin	Smooth and firm.	Smooth and firm with slight give on gentle pressure.
Stem	Firm and green.	May show early colour change.
Days from transplant	Approximately 60 to 70 days	Approximately 80 to 90 days

13.3 Harvesting Procedure

- The Agronomist must confirm harvest readiness before any fruit is picked.
- Use clean sharp scissors or a harvest knife only. Fruits must never be pulled from the plant by hand as this damages the stem and the plant.
- Cut the fruit stem cleanly, leaving 1 to 2 cm of stem attached to the fruit.
- Handle fruits with care. Do not drop, compress, or stack carelessly.
- Place harvested fruits immediately into clean food-grade harvest crates or buckets. Fruits must not be placed on the walkway, mulch film surface, or bare ground at any point.

13.4 Fruits Not to be Harvested

Do not harvest any fruit presenting:

- Sunken or dark spots
- Cracks or splits
- Soft or mushy texture
- Insect damage or entry holes
- Colour inconsistent with normal ripening

Set aside all substandard fruits for Agronomist review before disposal.

13.5 Harvest Frequency

- Once fruits begin to ripen, harvest must occur every 2 to 3 days.
- Overripe fruits left on the plant divert energy from developing fruits and create disease entry points.
- The Agronomist sets the harvest schedule based on crop stage and market requirements.

13.6 Post-Harvest Handling

- Move all harvested produce to the designated post-harvest area immediately. Do not hold produce inside the greenhouse after harvesting.
- Grade produce into three categories: Grade A (full size, no blemish), Grade B (minor imperfection, marketable), Reject (damaged, diseased, or substandard).
- Record all quantities by grade on Form E.
- Produce must not be washed with untreated or contaminated water.

13.7 End of Cycle Indicators

The Agronomist determines when the cycle is ending. Standard indicators include:

- Significant reduction in new flower and fruit formation
- Rapid decline in overall plant vigour
- Disease pressure that is no longer economically viable to manage
- Completion of the target cycle duration. Bell pepper under greenhouse conditions typically runs 6 to 9 months from transplanting.

SECTION 14 - CLEANING AND SANITATION

14.1 Daily Cleaning

- Remove all fallen leaves, pruned material, and dropped fruits from walkways and the mulch film surface after every session. Plant debris must not remain inside the greenhouse overnight.
- Wipe any surface or tool that has contacted diseased plant material using Hypo solution (50ml to 75ml Hypo in 10 to 15 litres of clean water).
- Maintain clear and unobstructed walkways at all times.
- Collect and remove any soil or growing medium displaced from the planting beds immediately.

14.2 Weekly Cleaning - Every Saturday

The full weekly cleaning of all active kits is carried out every Saturday. The Agronomist conducts a formal inspection after cleaning is complete and signs Form C.

Weekly cleaning tasks:

- Remove all plant debris from the entire kit including walkways, between plants along the beds, and along the base of the greenhouse structure.
- Inspect and clean drip line filters. Flush all drip lines with clean water.
- Wipe down the greenhouse structure including posts, support wires, and overhead frames with a damp cloth to remove dust, residue, and debris.
- Clean and disinfect all tools: scissors, harvest knives, stakes, and watering equipment using Hypo solution.
- Inspect the shade netting, side polythene, and roof covering. Report any tears, gaps, or damage to the Farm Manager.
- Clean and inspect the greenhouse entrance including the footwear station and handwashing point.
- Inspect and clean the gravelled drainage channel along the exterior base of the greenhouse wall. Remove any debris, silt, or vegetation obstructing drainage.
- Inspect the perimeter clearance zone. Report any grass or debris accumulation exceeding the standard in Section 2.1 to the Farm Manager.
- Inspect all mulch film along every bed. Record and report all damage.

14.3 Tool Disinfection Protocol

- Cutting tools must be disinfected between plants whenever working in any area with active pest or disease pressure.
- Disinfection solution: Hypo (sodium hypochlorite) - 1 part Hypo concentrate to 9 parts water. Approved products include Hypo, Jik, or any sodium hypochlorite-based product.
- Submerge the tool for a minimum of 30 seconds and allow to air dry before use on the next plant.
- A tool used on a diseased or suspect plant must not be used on a healthy plant without completing this disinfection step.

14.4 End of Crop Cycle Cleanout

At the end of every crop cycle, the following must be completed before any new crop is introduced:

- Remove all plants, stakes, drip lines, emitters, and all mulch film from the kit entirely.
- Clear and turn all planting beds. Remove all root material and crop residue from the soil.
- Scrub all internal surfaces, posts, wires, and frames with detergent and water.
- Apply full disinfection of all surfaces using Copper Oxychloride or Formaldehyde solution as directed by the Agronomist.
- Allow the greenhouse to rest for a minimum of 14 days before the next crop cycle begins.
- Flush all drip lines with dilute Hypo solution and rinse thoroughly with clean water before reinstalling for the next cycle.
- Install fresh mulch film on all beds before transplanting begins.

14.5 Waste Disposal

- All diseased plant material must be placed in sealed bags and removed from the farm entirely. It must not be composted, left on farm land, or placed near any greenhouse structure.
- Empty chemical containers must be disposed of safely as directed by the Agronomist. Containers must never be burned or reused for any other purpose.

SECTION 15 - TECHNICAL STANDARDS

15.1 Recommended Environmental Parameters

Table 15.1 - Environmental Parameter Standards

Parameter	Target Range	Response if Outside Range
Daytime Temperature	24 to 29 degrees Celsius	Above 32: manage open ends of the greenhouse immediately. Blossom drop risk increases sharply. Below 18: close open ends. Risk of chilling damage.
Night Temperature	18 to 22 degrees Celsius	Below 15: report to Farm Manager immediately. Chilling damage and growth inhibition risk.
Relative Humidity	60 to 75 percent	Above 80: increase air movement through open ends immediately. Fungal disease risk rises sharply. Below 50: monitor for water stress.
pH of Nutrient Solution	5.8 to 6.5	Below 5.8: toxicity risk from manganese and iron. Above 6.5: iron, manganese, and zinc become unavailable to the plant.
EC of Nutrient Solution	2.0 to 3.5 mS/cm	Below 2.0: solution too dilute, plants underfed. Above 3.5: osmotic stress, root burn, wilting despite irrigation.

15.2 Nutrient Programme by Growth Stage

Table 15.2 - Nutrient Programme by Growth Stage

Growth Stage	Nutrient Focus	Target EC	Notes
Establishment (Week 1 to 2)	Phosphorus dominant	1.5 to 2.0 mS/cm	Promotes root development. Keep nitrogen low.
Vegetative (Week 3 to 6)	Nitrogen dominant	2.0 to 2.5 mS/cm	Drives leaf and stem growth. Monitor for excessive vegetative growth.
Flowering (Week 7 to 10)	Increase Potassium and Calcium	2.5 to 3.0 mS/cm	Reduce nitrogen. Supports flower development. Prevents blossom end rot.
Fruit Set and Development (Week 10 to 14)	Potassium dominant, sustained Calcium	2.5 to 3.5 mS/cm	Supports fruit size, colour, and shelf life. Maintain calcium throughout.
Harvest Period	Balanced, maintain K and Ca	2.0 to 3.0 mS/cm	Do not reduce nutrients during active harvest. Monitor plant vigour.

15.3 Nutrient Deficiency Identification

Table 15.3 - Nutrient Deficiency Visual Guide

Nutrient	Visual Symptom	First Appears On	Urgency
Nitrogen (N)	General yellowing of entire leaves. Pale overall appearance. Slow growth.	Older lower leaves	Same day
Phosphorus (P)	Purple or reddish colouration on underside of leaves.	Older leaves	Same day
Potassium (K)	Brown scorching at leaf edges and tips.	Older leaves	Same day

Nutrient	Visual Symptom	First Appears On	Urgency
Calcium (Ca)	Blossom end rot on fruits. Tip burn on young leaves.	Young leaves and fruits	Immediately
Magnesium (Mg)	Interveinal chlorosis - yellowing between veins, veins remain green.	Older leaves	Same day
Iron (Fe)	Interveinal chlorosis on young leaves only. Veins remain green.	Young new leaves	Same day
Boron (B)	Distorted, thickened, or brittle new growth. Hollow or corky fruit interior.	Young growth and fruits	Immediately

15.4 Pollination Management

- Bell pepper in a closed greenhouse has restricted natural pollination due to limited insect access.
- Mechanical pollination is achieved by gentle vibration of the overhead support wires or by lightly tapping flowering stems during mid-morning between 9:00 AM and 11:00 AM.
- If fruit set rate falls below 50 percent of observed flowers, the Agronomist reviews temperature, humidity, and nutrient balance before considering additional intervention.
- Workers must not touch flowers or flowering stems without explicit Agronomist instruction.

SECTION 16 - FARM MANAGER OVERSIGHT

16.1 Daily Responsibilities

- Collect Form A from the Agronomist every evening. Review all entries.
- Verify that Notion records have been updated before 7:00 PM daily.
- Follow up on all flagged observations, pest or disease reports, irrigation faults, or missed records before the following morning.
- Confirm all greenhouse kits are secured at the end of each day during evening rounds.

16.2 Weekly Inspection - Every Saturday

The Farm Manager conducts a formal inspection of all active greenhouse kits every Saturday alongside the Agronomist.

Table 16.1 - Weekly Greenhouse Inspection Checklist

Area	What to Check	Acceptable Standard
Plant health - all kits	Colour, uniformity, vigour	No widespread yellowing, wilting, or unmanaged pest or disease activity
Drip lines and emitters	All functioning, no leaks	Every emitter delivering. No pooling or dry sections on beds.
Header tank level	Adequate water for next run	Minimum half capacity
pH and EC records on Notion	Logged before every run	All entries current and complete
Chemical store	Labelled, locked, organised	No unlabelled containers. Store secured.
Greenhouse structure - all kits	Shade netting, polythene, frames	No gaps, tears, or structural damage
Mulch film - all beds	Condition and integrity	No tears, lifting, or displaced sections
Cleanliness - all kits	Walkways, bed surfaces, structure	No debris. No standing water. Walkways clear.
Perimeter clearance zone	Grass height and debris	Grass below 5cm. No debris. No materials within 2 metres.
Drainage channel	Exterior base of greenhouse wall	Clear of debris, silt, and vegetation
Nursery area or kit	Seedling health, tray condition	No damping off. Trays elevated. No failed seedlings remaining.
Binding wire and stakes	All plants supported	No fallen or unsupported plants. No tight bindings on stems.
Harvested produce	Grading and recording	Produce graded and recorded correctly on Form E.
Form A completeness	All entries filled and signed	No missing entries or unsigned days.
Door discipline	Doors closed during all operations	No kit door left open at any time during operations.
Foliar spray records	All applications on Form D	Every foliar application recorded with product and rate.

16.3

The Farm Manager signs Form C every Saturday after the inspection is complete.

16.4

The Farm Manager approves all chemical and foliar fertiliser purchases before procurement. No product enters the farm without Farm Manager authorisation.

16.5

The Farm Manager does not override Agronomist crop management decisions without consultation. Unresolved disagreements are escalated to the CEO.

SECTION 17 - NOTION RECORDS

17.1

The Agronomist updates the farm Notion system daily before 7:00 PM with the following parameters for each active kit:

- Morning and evening pH readings
- Morning and evening EC readings
- Any pH or EC adjustments made including additive used and volume
- Irrigation run times and duration
- Pump run details
- Foliar applications with product name, formulation, dilution rate, and rows treated
- Pest or disease observations and actions taken
- Plant losses
- Chemical applications with product name, rate, and rows treated
- Any escalations to the Farm Manager

17.2

Physical forms (Form A through Form E) are the field record of all daily operations and must be completed in full before Notion entry. Notion is the digital system of record. Both must be maintained without exception.

17.3

The Farm Manager reviews Notion records daily and confirms completeness. Any gap in the Notion record is treated as a compliance failure.

SECTION 18 - SANCTION GRID

Table 18.1 - Disciplinary Sanction Grid

Breach	First Offence	Second Offence	Third Offence
Entering greenhouse without following entry protocol	Verbal warning. Recorded.	Written warning.	Suspension from greenhouse unit.
Leaving the greenhouse door open while inside	Verbal warning. Recorded.	Written warning.	Suspension.
Taking equipment between kits without cleaning and disinfection	Verbal warning. Recorded.	Written warning.	Suspension.
Applying any chemical without Agronomist authorisation	Suspension.	Termination.	N/A
Applying foliar feed without Agronomist instruction	Verbal warning. Recorded.	Written warning.	Suspension.
Removing, pruning, or adjusting any plant without instruction	Verbal warning. Recorded.	Written warning.	Suspension.
Leaving plant debris on walkways or mulch film surface after session	Verbal warning. Recorded.	Written warning.	Pay deduction.
Running pump unattended	Written warning.	Suspension.	Termination.
Failing to report a pest or disease observation same day	Written warning.	Suspension.	Termination.
Harvesting without Agronomist confirmation	Written warning.	Suspension.	Termination.
Tying binding wire tightly against the plant stem	Verbal warning. Recorded.	Written warning.	Suspension.
Form A not completed or submitted same day	Verbal warning. Recorded.	Written warning.	Pay deduction for the day.
Unauthorised person permitted inside any kit	Written warning.	Suspension.	Termination.
Chemical containers not returned to locked store after use	Written warning.	Suspension.	Termination.
Agronomist failing to check pH and EC before fertigation run	Written warning.	Written report to Farm Manager.	Escalation to CEO.
Agronomist failing to update Notion records before 7:00 PM	Verbal warning. Recorded.	Written warning.	Escalation to Farm Manager.
Farm Manager failing to conduct Saturday weekly inspection	Verbal warning from CEO. Recorded.	Written report to CEO.	N/A

FORMS AND RECORDS

FORM A - WEEKLY DAILY ENTRY FORM

WGNI FARM - GREENHOUSE WEEKLY DAILY ENTRY FORM

Kit Number	Crop	Growth Stage	Week of (Mon - Sun)
_____	Bell Pepper	_____	_____ to _____

Worker Name: _____

MONDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

TUESDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

WEDNESDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		

Item	Morning Entry	Evening Entry
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

THURSDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		

Item	Morning Entry	Evening Entry
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

FRIDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

SATURDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		

Item	Morning Entry	Evening Entry
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____

Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

SUNDAY Date: _____

Item	Morning Entry	Evening Entry
Time of Entry		
Header Tank Level (Full / Half / Low)		
Pump Run Done (Yes / No)		
Pump Start Time		
Pump Stop Time		
pH of Nutrient Solution (Agronomist)		
EC of Nutrient Solution (Agronomist)		
pH Adjusted (Yes / No)		
EC Adjusted (Yes / No)		
Fertigation Start Time		
Fertigation End Time		
Nutrient / Foliar Solution Details		

Item	Morning Entry	Evening Entry
Drip Line Faults (Yes / No - describe)		
Plants with Visible Problems (Yes / No - describe)		
Pest or Disease Signs (Yes / No - describe)		
Mulch Film Damage (Yes / No - describe)		
Plant Debris Removed (Yes / No)		
Door Secured on Entry (Yes / No)		
Any Other Observation		

Worker Sig (AM): _____ Worker Sig (PM): _____
 Agronomist Signature: _____ Time Reviewed: _____ Notion Updated: Yes / No

WEEKLY SUMMARY

Item	Total for the Week
Total Pump Runs This Week	
Total Plants Lost This Week	
Total Chemical Applications This Week	
Total Foliar Applications This Week	
Total Harvests This Week	
Any Escalations to Farm Manager (Yes / No)	

Farm Manager Final Weekly Signature: _____ **Date (Sunday):** _____
 Week Status: Satisfactory / Unsatisfactory (circle one)
 Farm Manager Comments: _____

FORM B - pH AND EC DAILY LOG**WGNI FARM - pH AND EC DAILY LOG**

Kit Number	Week of
_____	_____

Date	Morning pH	Morning EC	Adjusted (Yes/No)	Evening pH	Evening EC	Adjusted (Yes/No)	Agronomist Signature
Monday							
Tuesday							
Wednesday							
Thursday							
Friday							
Saturday							
Sunday							

Target pH: 5.8 to 6.5 Target EC: 2.0 to 3.5 mS/cm**Farm Manager Review Signature: _____ Date: _____**

FORM C - WEEKLY CLEANING SIGN-OFF FORM**WGNI FARM - GREENHOUSE WEEKLY CLEANING SIGN-OFF**

Kit Number	Date of Inspection	Worker Who Cleaned
_____	_____ (Every Saturday)	_____

Area	Standard	Result	Action Required
Walkways	Clear and unobstructed, no debris	Pass / Fail	
Planting Bed Surfaces	No debris or fallen fruits on mulch film	Pass / Fail	
Mulch Film Integrity	No tears, lifting, or displacement	Pass / Fail	
Drip Lines and Filters	Flushed and clean	Pass / Fail	
Tools and Equipment	Cleaned and disinfected with Hypo solution	Pass / Fail	
Structure (Posts, Wires, Frames)	Wiped down, no residue or debris	Pass / Fail	
Shade Netting	No tears or sagging sections	Pass / Fail	
Side Polythene and Roof Covering	No damage or gaps	Pass / Fail	
Chemical Store	Locked, labelled, organised	Pass / Fail	
Waste Removal	All debris removed from greenhouse	Pass / Fail	
Perimeter Zone	Grass below 5cm, no debris or materials	Pass / Fail	
Drainage Channel	Clear of debris, silt, and vegetation	Pass / Fail	
Greenhouse Entrance and Footwear Station	Clean and organised	Pass / Fail	

Overall Result	Areas to Redo
Pass / Fail (circle one)	

Re-inspection Completed	Date	Time
Yes / No	_____	_____

Agronomist Signature: _____ Date: _____

Farm Manager Signature: _____ Date: _____ Time: _____

FORM D - CHEMICAL AND FOLIAR APPLICATION RECORD**WGNI FARM - CHEMICAL AND FOLIAR APPLICATION RECORD**

Kit Number	Crop Stage	Date	Applied By
_____	_____	_____	_____

Item	Details
Type of Application	Pesticide / Fungicide / Foliar Feed (circle one)
Pest, Disease, or Deficiency Being Treated	
Product Name	
Active Ingredient or Nutrient Formulation	
Previous Product Used for Same Pest (rotation check)	
Dilution Rate	
Volume Prepared	
Method of Application (Foliar Spray / Soil Drench)	
Rows or Areas Treated	
Time of Application	
Re-entry Interval (from product label) - pesticide only	
Re-entry Permitted From (date and time) - pesticide only	
Protective Gear Worn (Yes / No)	
Product Returned to Store After Use (Yes / No)	

Worker Signature: _____ Date: _____

Agronomist Authorisation Signature: _____ Date: _____

Farm Manager Awareness Signature: _____ **Date:** _____

FORM E - HARVEST RECORD FORM**WGNI FARM - HARVEST RECORD FORM**

Kit Number	Harvest Date	Harvest Number This Cycle	Harvested By
_____	_____	_____	_____

Grade	Quantity (kg)	Number of Fruits	Destination
Grade A (Full size, no blemish)			
Grade B (Minor imperfection, marketable)			
Reject (Damaged, diseased, substandard)			
TOTAL			

Reject Disposal Method: _____

Observations During Harvest: _____

Confirmed By (Agronomist): _____

Worker Signature: _____ Date: _____

Agronomist Signature: _____ **Date:** _____

DOCUMENT AUTHORISATION

FIELD	DETAIL
Document Title	Greenhouse Operations Standard Operating Procedure
Organisation	WGNI Farm, Abeokuta, Nigeria
Version	1.0
Date Issued	April 2026
Prepared By	Farm Manager / Agronomist
Authorised By	CEO, WGNI Farm
Next Review Date	October 2026

This SOP is authorised by the CEO of WGNI Farm. It covers all active greenhouse kits. Every new kit commissioned automatically falls under the full scope of this document without requiring a separate SOP. This document is subject to periodic review. All personnel are expected to know and comply with every section relevant to their role.